MIL-I-14770A(PA)
9 June 1975
SUPERSEDING
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17 March 1969

MILITARY SPECIFICATION

INDICATOR, REMOTE: M1

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of indicating device entitled as Indicator, Remote which is used on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-5350	- Transmitter, Synchro, Aircraft, General Specification For.
MIL-L-25142	- Luminescent Flourescent Material.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

PICATINNY ARSENAL

8785476 - Remote Indicator.

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8808362 - Packing and Marking for Storage and Shipments.
8863176 - Carton, Packing, Ammunition for Indicator, Remote Control.
EL-8785476 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The indicator shall comply with all requirements specified on the drawing (dwg) 8785476, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 General requirements.

3.5.1 Counters. Two sets of four-wheel, drum-type counters shall be used: one set for positive indications and one set for negative indications. The counters shall read in normal digital order. The units and tens shall be combined on one drum and shall be marked in numerically increasing sequence in increments of ten from "00" through "90" to "00". The unit "0" shall always be displaced with each ten's digit. The hundreds, thousands and tens of thousands wheels shall each be marked in numerically-increasing sequence from "0" through "9" to "0". The units-tens wheel may move continuously while the remaining wheels shall step from position to position. The movements shall be smooth and uniform without observable jitter at any time. Positive digits shall appear to move downward and negative digits shall appear to move upward when the indication is increasing numerically.

3.5.2 Dial and numeral markings. All markings shall be green on black except the tens of thousands and thousands wheels which shall be black on green. A comma shall be marked on the dial face between the hundreds and thousands wheels, and a plus type sign (+) and a minus type sign (-) to the left and in line with the respective tens of thousands wheel. The

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green numerals, dial markings, and the green background of the thousands and tens of thousands wheels shall be coated with fluorescent luminescent material conforming to MIL-L-25142 or approved equivalent.

3.5.3 Shutters. The counter not in use shall be covered by a shutter except at zero feet when both counters shall be visible. The shutters shall be activated by a 10-foot change from zero.

3.5.4 Stops. Internal stops shall be provided at both ends of the travel to prevent damage to the instruments.

3.5.5 Operating power. The range of operating power for the instrument shall be 26 ± 2 volts alternating current root mean square (vacrms), 400 ± 20 Hertz (Hz), single phase, isolated from instrument ground.

3.5.6 Transmitter. The indicator operation shall be compatible with synchro transmitters with electrical characteristics as specified in MIL-T-5350. Where dual transmitters are used, the transmitters shall be connected through a 20 to 1 gear ratio similar to Figure 1. The coarse transmitter must be within ± 1 degree of electrical when the fine transmitter is at electrical zero.

3.6 Insulation resistance. The indicator shall have a minimum insulation resistance of 5 megohms at 500 ± 50 volts direct current (vdc) between each pin and the indicator case when tested as specified in 4.4.1.

3.7 Electrical zero. The zero counter indication shall be set to agree with the electrical zero of the remote indicator when tested as specified in 4.4.2.

3.8 Operation. The indicator shall follow a signal from a synchro transmitter similar to the one described in Figure 1. Both synchro transmitters shall be at electrical zero when the angle measuring device indicates zero when tested as specified in 4.4.3.

3.9 Accuracy. The indicator reading shall have an accuracy of ± 10 feet at any setting within the range of -3,600 to 31,400 feet when tested as specified in 4.4.4.

3.10 Range. The indicator shall have a usable positive range from zero to 31,400 feet and a usable negative range from zero to -3,600 feet. Testing shall be as specified in 4.4.5.

3.11 Environmental testing. The indicator shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The indicator shall also comply with 3.6 through 3.10 before and after each environmental test. Testing shall be as specified in 4.4.6. Tests shall be conducted in the order specified.

3.11.1 High temperature. The indicator shall withstand exposure to a temperature of 165 ± 5 degrees Fahrenheit (°F) without evidence of damage or impairment of operation and shall comply with the requirements of 3.6 through 3.10 as specified in 4.4.6.1.

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3.11.2 Low temperature. The indicator shall withstand exposure to a temperature of -65 ± 5°F and after stabilization, (see 6.4) shall comply with the requirements specified in 3.6 through 3.10 when tested as specified in 4.4.6.2

3.11.3 Vibration. The indicator shall withstand vibrations throughout a frequency range of 10-55-10 Hz at a double amplitude of 0.064 inch. There shall be no evidence of damage or impairment of operation and the remote indicator shall comply with the requirements of 3.6 through 3.10 when tested as specified in 4.4.6.3.

3.11.4 Humidity. The indicator shall withstand exposure to a relative himidity of 90 to 98 percent and an elevated temperature without evidence of damage or impairment of operation. Upon completion of this test, the indicator shall comply with the requirements of 3.6 through 3.10 when tested as specified in 4.4.6.4.

3.11.5 Limited operating conditions. The indicator shall be capable of operating for 10 cycles of the range test (see 3.10) during the following conditions without loss of accuracy or degradation. One range test cycle shall consist of the remote indicator indicating from 0 feet to -3,600 feet through to 31,400 feet and return to 0 feet. Upon the completion of the 10 cycles at each environment, the remote indicator shall comply with 3.6 through 3.10. Testing shall be as specified in 4.4.6.4.

3.11.5.1 High temperature. The indicator shall be operated at a temperature of +130 ± 5°F, with a minimum input power of 30 volts ac, minimum frequency of 440 Hz and a minimum follow rate of 250 feet per second.

3.11.5.2 Low temperature. The indicator shall be operated at a temperature of -40 ± 5°F, with a maximum imput power of 26 volts ac, maximum frequency of 360 Hz and a minimum follow rate of 150 feet per second.

3.12 Life cycle. The indicator shall be operated continuously for 10,000 cycles from -3,600 feet to 31,400 feet and back to -3,600 feet at a minimum follow rate of 250 feet per second without any degradation of performance or accuracy. Testing shall be as specified in 4.4.7.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 8785476) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Insulation resistance (see 3.6)		X
3. Electrical zero (see 3.7)		X
4. Operation (see 3.8)		X
5. Accuracy (see 3.9)		X
6. Range (see 3.10)		X
7. High temperature (see 3.11.1)		X
8. Low temperature (see 3.11.2)		X
*9. Vibration (see 3.11.3)	X	
*10. Humidity (see 3.11.4)	X	
*11. Limited operating conditions (see 3.11.5)		X
*12. Life cycle (see 3.12)		X

*One unit selected at random from the first article sample.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Remote Indicator assembly (see dwg 8785476).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Marking is missing, incomplete, incorrect, or unidentifiable	Visual
	102. Housing has separations, broken glass, breaks, cracks, dents, or chips	Visual
	103. Improper use and locations of luminescence on numbers and dial markings	Visual
	104. Incorrect locations of polarizing key	Visual
	105. Mounting nuts missing or insecure (4)	Visual/Manual
	106. Looseness of electrical receptacle	Visual/Manual
	107. Pins of receptacle broken or loose	Visual/Manual
	108. Insecurity of bezel to indicator housing	Visual/Manual
	109. Objective evidence (see 6.2) not available for transmitters in accordance with MIL-T-5350	Visual
	110. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Protective coating inadequate	Visual
	202. Presence of corrosion	Visual
	203. Name plate is missing, incomplete, incorrect, or unidentifiable	Visual

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204. Evidence of poor workmanship (see 3.13).....Visual

4.3.2.1.2 Carton, packing, ammunition for indicator, remote control (see dwg 8863176 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Material improper (see 3.1)	Visual
	102. Bag length before packing, min	*Measure
	103. Bag width before packing, min	*Measure
	104. Objective evidence (see 6.2) not available for heat seal test in accordance with MIL-P-116	Visual
	105. Incorrectly packed	Visual/Manual
	106. Any tear, split, or crack in bag	Visual

*Measure with suitable scale.

Minor:		
	201. Marking incorrect or improper	Visual
	202. Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.3 Packing and marking for storage and shipments (see dwg 8808362 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Contents not tight in package	Manual
	102. Incorrectly packed	Visual/Manual
	103. Package material improper (see 3.1)	Visual
Minor:		
	201. Marking incorrect or improper	Visual
	202. Evidence of poor workmanship (see 3.13)	Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Insulation resistance	Major
b. Electrical zero	Major
c. Operation	Major

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d.	Accuracy	Major
e.	Range	Major
f.	High temperature	Major
g.	Low temperature	Major
h.	Vibration	Major
i.	Humidity	Major

Of these tests, h and i are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Insulation resistance, electrical zero, operation, accuracy and range. Each indicator shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.5 through 3.10, the item shall be rejected and removed from the lot.

4.3.3.2 High temperature, low temperature, vibration and humidity. Two sample indicators shall be selected from every 15 indicators or portion thereof, produced. The sample shall be tested in accordance with 4.4.6.1 through 4.4.6.4. If any item fails to comply with 3.11.1 through 3.11.4, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8785476). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. Apply 500 ± 50 vdc between each pin and the remote indicator case. The resistance reading shall be stable and not less than 5 megohms.

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4.4.2 Electrical zero test. Each remote indicator shall be tested as specified in the following paragraphs and as required in 3.7.

4.4.2.1 Test circuit and interconnections. Connect the indicator to an electrical zero test set constructed in accordance with the schematic of Figure 2 and the following: The standard synchro transmitter shall have a secondary open circuit voltage of 11.8 ± 0.3 vac and shall have its shaft connected to a precision angle measuring device capable of being read to an accuracy of ± 0.1 degree. This synchro transmitter shall be set to electrical zero when the precision angle measuring device is set to zero degree. The electrical zero test also shall include an electrical zero transformer connected as shown in Figure 2 which will have an open circuit voltage of 11.8 ± 0.3 vac. A power switch supplying 26 ± 0.5 vac, 400 ± 10 Hz power shall be connected as shown. A three-position "TEST" switch connected as shown shall have a "CAL" "ZERO SET" and "TEST" position. It would be preferable if the "CAL" position were momentary in order to prevent random readings which could occur if the remote indicator were connected first to the "CAL" position.

4.4.2.2 Step 1: Indicator electrical zero test. With the "PWR" switch in its "OFF" position, set the "TEST" switch to its "ZERO SET" position. Turn "PWR" switch to its "ON" position. The indicator, if not at zero indication, shall drive to zero ± 10 feet indication. No jitter or drift of setting shall be apparent.

4.4.2.3 Step 2: Standard synchro transmitter calibration check. This step is a cross check of the electrical zero transformer and the synchro transmitter of the electrical zero test set. Align the synchro transmitter to zero degree to correspond with the zero degree on the precision angle measuring device. Switch the test switch to its "CAL" position. The indicator shall indicate the same setting as was indicated under Step 1.

4.4.2.4 Step 3: Electrical zero check. When step 2 above is satisfactory, proceed with the following: Turn the test switch to its "TEST" position. Rotate the standard synchro transmitter clockwise through the range of 1 degree ± 0.1 degree of the precision angle measuring device from zero degrees. The counters of the remote indicator may change as the 1 degree change is made, but once the counter assumes a new setting, the counter shall not change more than ± 10 feet nor shall it display any tendency to drift or jitter. Next, rotate the transmitter counter clockwise back to zero degree. Continue this rotation in a counter clockwise direction through a range of -1 degree ± 0.1 degree of the precision angle measuring device from zero degree. The counters of the remote indicator may change as the -1 degree change is made, but once the counter assumes a new setting, the counter shall

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not change more than ± 10 feet nor shall it display any tendency to drift or jitter.

4.4.3 Operational tests. The indicator shall be tested as specified in the following paragraphs.

4.4.3.1 Test circuit and interconnection. Connect the indicator to a transmitting device constructed in accordance with the schematic of Figure 1 and the following: A standard synchro transmitter having a secondary open circuit voltage of 11.8 ± 0.3 vac shall have its shaft connected to a precision angle-measuring device capable of being read to an accuracy of ± 0.1 degree. A second transmitter shall be geared to the shaft of the first in such a way that the second transmitter will make one complete revolution (360 electrical and mechanical degrees) for 20 complete revolutions (7200 electrical and mechanical degrees) of the first. Both transmitters shall be set to electrical zero when the angle measuring device is set to zero degrees. Electrical connections shall be brought out to the transmitter as shown in Figure 1 with 26 ± 0.5 vac, 400 ± 10 Hz power supplied as shown. A switch shall be provided to turn this power on and off. It should be noted that during the following functional checks, there shall be no observable jitter of the counters at null, or while the indications are increasing or decreasing. If desired, the standard transmitter may be calibrated in feet instead of degrees corresponding to the indications of the remote indicator.

4.4.3.2 Test at null. Apply power and allow the indicator to come to null. Rotate the standard transmitter until the indicator shown zero feet. Both counters shall be visible and readable at this time, and the standard transmitter setting shall be 0 degrees ± 1 degree.

4.4.3.3 Test at ccw 1 degree. Slowly rotate the standard transmitter counterclockwise until minus 10 feet is indicated by the indicator. At this point, the upper set of digits of the indicator shall be covered by the shutter, and the standard transmitter setting shall be -1 degree ± 1 degree. Return remote indicator to conditions established by performing test 4.4.3.2.

4.4.3.4 Test at cw 1 degree. Slowly rotate the standard transmitter clockwise until plus 10 feet is indicated by the indicator. At this point, the lower set of digits shall be covered by the shutter, and the standard transmitter setting shall be +1 degree ± 1 degree. Return indicator to conditions established by performing test 4.4.3.2.

4.4.4 Test of accuracy. Using the test circuit and interconnections specified in 4.4.3.1, test the indicator as follows: With the standard transmitter set at zero degrees and the indicator indicating zero ± 10 feet when power is on, perform the following test sequentially:

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Turn Power	Rotate Fine Transmitter	Turn Power	Indicator Reading (Feet)
Off	360°ccw	On.....	-3600 ± 10
Off	10° cw	On.....	-3500 ± 10
Off	350° cw	On.....	0 ± 10
Off	90° cw	On.....	900 ± 10
Off	90° cw	On.....	1800 ± 10
Off	90° cw	On.....	2700 ± 10
Off	90° cw	On.....	3600 ± 10
Off	360° cw	On.....	7200 ± 10
Off	360° cw	On.....	10,800 ± 10
Off	360° cw	On.....	14,400 ± 10
Off	360° cw	On.....	18,000 ± 10
Off	360° cw	On.....	21,600 ± 10
Off	360° cw	On.....	25,200 ± 10
Off	360° cw	On.....	28,800 ± 10
Off	360° cw	On.....	31,400 ± 10
Off	360°ccw	On.....	28,800 ± 10

Upon completion of above test, turn power off. With power off, rotate the standard transmitter counterclockwise 2880 degrees (eight turns). Turn power on and measure the time interval which elapses as the indicator returns to null. The indicator shall read zero ± 10 feet at null and the measured time intervals shall not exceed 115 seconds. There shall be no observable jitter at any time. The follow rate shall be 250 feet per second minimum.

4.4.5 Range tests. Using the test circuit and interconnections specified in 4.4.3.1, the indicator shall be tested as specified below.

4.4.5.1 Test of negative stop. Turn power on. Set transmitter at zero degrees and the indicator indicating zero ± 10 feet. Turn power off. The transmitter to 540 degrees ccw. Turn power on. The indicator shall drive in a negative direction to a value between -3600 and -5000 feet; then the direction of indication change shall reverse causing decreasing negative indicator values to appear. This shall continue until at some value between -3600 feet and -4600 feet the direction of rotation shall again reverse. The cycling shall continue within those limits as long as the setting conditions remain unchanged. Turn power off.

4.4.5.2 Test of positive stop. Turn power on. Set transmitter at zero degrees and the indicator indicating zero ± 10 feet. Turn power off. Rotate transmitter 3320 degrees cw (9 turn plus 80 degrees). Turn power on. The remote indicator shall drive in a positive direction to a value between 31,400 feet and 32,800 feet whereupon the direction of indication change shall reverse causing decreasing indicator values to appear. This shall continue until at some value between 31,400 feet and 32,400 feet the direction of rotation shall again reverse. This cycling shall continue within these limits as long as the setting conditions remain unchanged. Turn power off.

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4.4.6 Environmental testing (see 3.11).

4.4.6.1 High temperature. The indicator with electrical receptacle covered shall be placed in a test chamber and stabilized (see 6.4) at a temperature of 165 ± 5°F with a maximum relative humidity of 15 percent for ten hours minimum. It shall then be stabilized to ambient room temperature. No forced cooling shall be employed such as directing air on the remote indicator by means of a fan or placing the indicator in refrigerated air. The indicator after stabilization shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4 and 4.4.5 and shall comply with 3.11.1.

4.4.6.2 Low temperature. The indicator with electrical receptacle covered shall be placed in test chamber and stabilized (see 6.4) at a temperature of -65 ± 5°F for ten hours minimum. It shall then be stabilized at ambient room temperature. No forced warming shall be employed, such as directing warm air on the indicator by means of a fan or placing the indicator in warm air. The indicator after stabilization shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4 and 4.4.5 and shall comply with 3.11.2.

4.4.6.3 Vibration. The indicator shall be secured to a vibration table and vibrated for 45 minutes minimum in each of three mutually perpendicular planes through a frequency range of 10 to 55 Hz to 10 Hz at a double amplitude of 0.064 inch. The frequency shall vary uniformly from 10 to 55 to 10 Hz making approximately one cycle per minute. Upon completion of the test the instrument shall be tested as specified in 4.4.1. 4.4.2, 4.4.3, 4.4.4 and 4.4.5 and shall comply with 3.11.3.

4.4.6.4 Limited operating conditions.

4.4.6.4.1 High temperature. The indicator, with power off, shall be placed in a test chamber and stabilized (see 6.4) for a minimum of 3 hours at a temperature 130 ± 5°F with a maximum relative humidity of 5 percent. While at this temperature the indicator shall be operated under conditions specified in 3.11.5.1, for 10 cycles of the range test as specified in 4.4.5. One range test cycle shall consist of the indicator indicating from 0 feet to -3,600 feet through to 31,400 feet and return to 0 feet. Upon the completion of the 10th cycle, the indicator shall then be allowed to return to room ambient conditions. No forced cooling shall be employed, such as, directing air on the indicator by means of a fan or placing the indicator in refrigerated air. The remote indicator, after stabilization, shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4 and 4.4.5.

4.4.6.4.2 Low temperature. The indicator, with power off, shall be placed in a test chamber and stabilized (see 6.4) for a minimum of 3 hours at a temperature of -40 ± 5°F with a relative humidity that will produce frozen condensate.

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While at this temperature the indicator shall be operated under conditions specified in 3.11.5.2 for 10 cycles of the range test specified in 4.4.5. One range test cycle shall consist of the indicator indicating from 0 feet to -3,600 feet through to 31,400 feet and return to 0 feet. Upon the completion of the 10th cycle, the indicator shall then be allowed to return to room ambient conditions. No forced cooling shall be employed, such as, directing air on the indicator by means of a fan or placing the remote indicator in warm air. The indicator, after stabilization, shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4 and 4.4.5.

4.4.7 Life cycle. The indicator shall be operated continuously for 10,000 cycles from -3,600 feet to 31,400 feet and back to -3,600 feet at a minimum follow rate of 250 feet per second utilizing approved measuring equipment. Upon the completion of this test, the remote indicator shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.4.5.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The indicator shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.1.2 Level C. The indicator shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Packing.

5.1.2.1 Levels A and C. The indicator shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The indicator shall be packaged in accordance with dwg 8863176 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The indicator shall be packaged in accordance with dwg 8863176 and 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The indicator shall be packed in accordance with dwg 8863176 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8836176 and 8808362.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

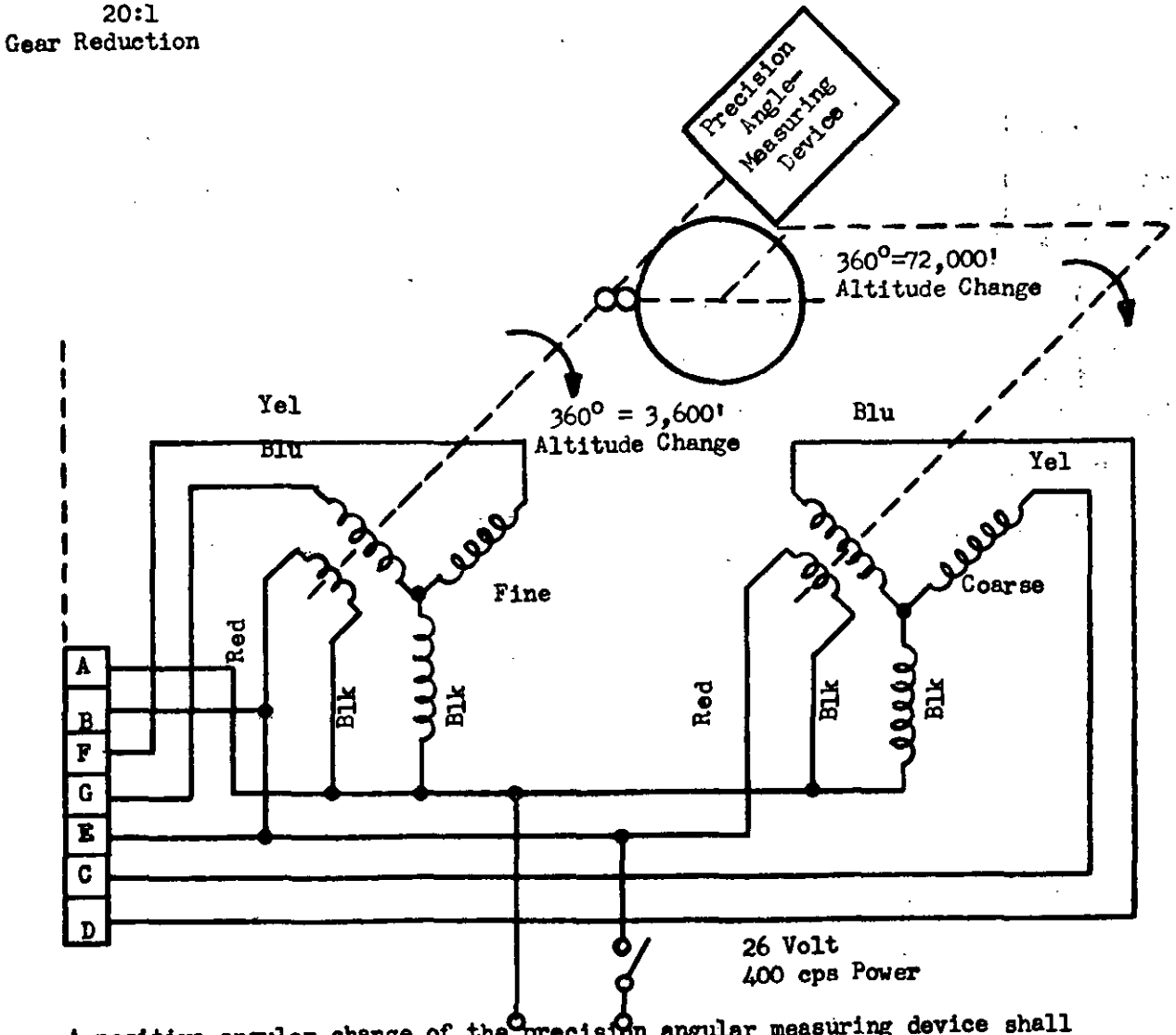
6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A093

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A positive angular change of the precision angular measuring device shall produce a clockwise rotation of the standard transmitter which, when connected to the Remote Indicator, shall cause an increase in the indicated setting of the Remote Indicator.

Figure 1. Suggested Test Circuit and Transmitter Schematic

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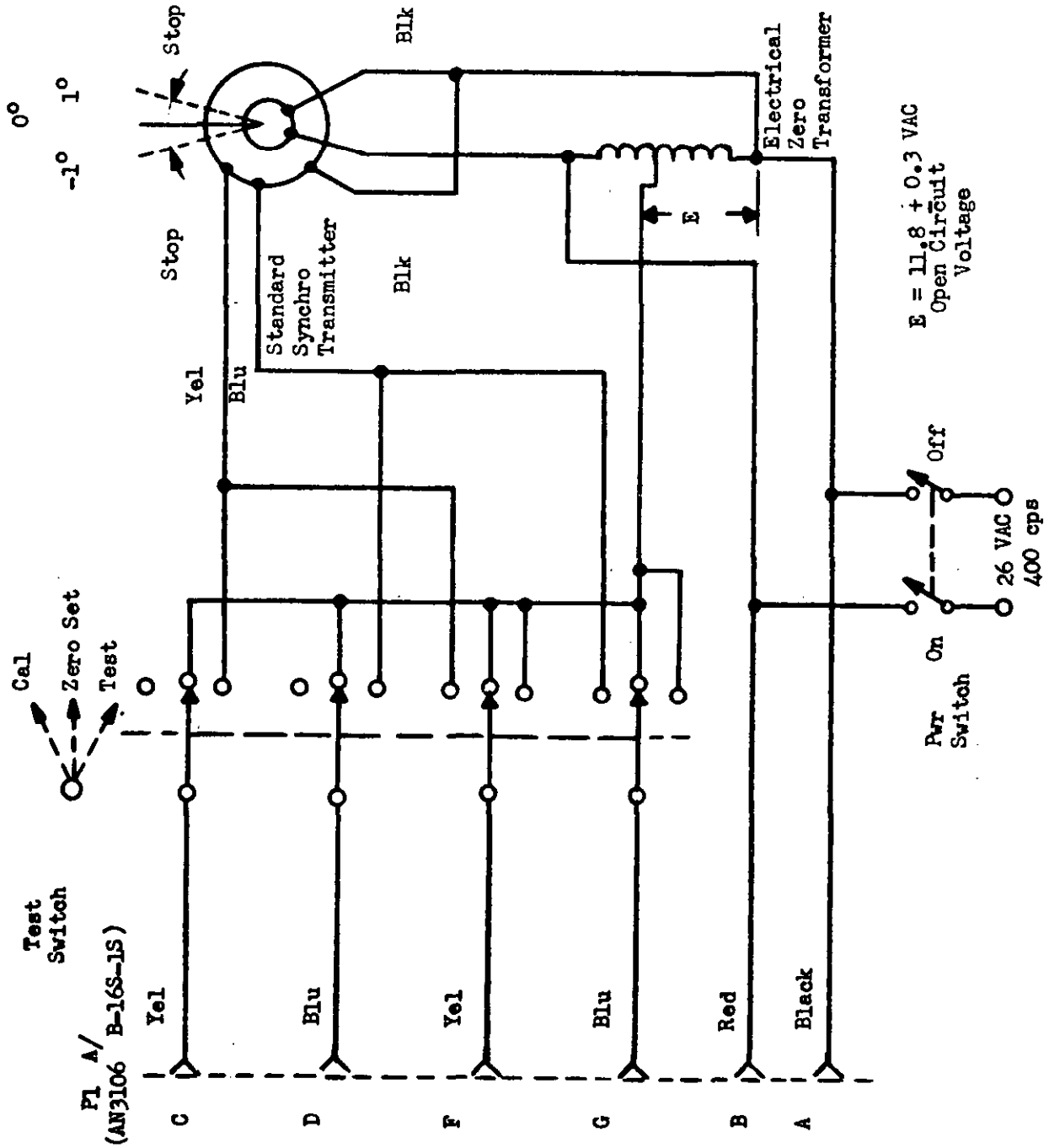


Figure 2. Suggested Electrical Zero Test Set

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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